

Indirect Retainers

Learning Objectives:

By the end of this lecture, students will be able to:

- Define and explain the function of indirect retainers in removable partial dentures.
- Describe distal extension base movement and the fulcrum line concept.
- Identify optimal indirect retainer placement and challenges with incisor placement.
- List and describe various indirect retainer forms.
- Evaluate the pros and cons of using rugae support for maxillary denture retention.

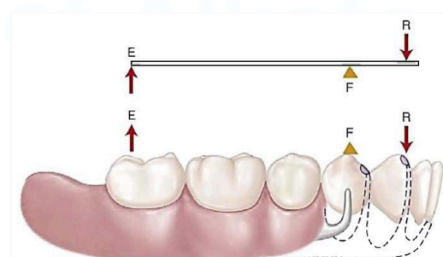
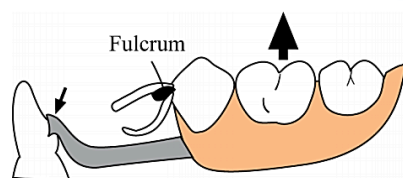
Indirect Retainer: A part of a removable partial denture that prevents rotational displacement of the denture about the rests of the principal abutment teeth. Indirect retainers usually take the form of rests on the opposite side of a fulcrum line.

Vertical movements of distal extension bases can occur in two directions:

1. Movement of a distal extension base toward the ridge tissues. This movement is proportional to tissue displacement, the denture base's fit and the load applied.
2. Movement of a distal extension base away from the ridge tissues. This movement occurs via either displacement of the entire denture (resisted by the direct retainers) or a rotational movement about an axis. This axis passes through the rests of the most posterior abutment teeth and is called the **fulcrum line**.

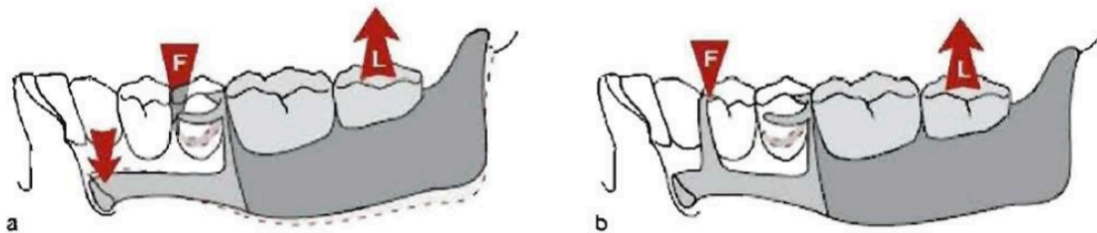
Rotation of a partial denture about the fulcrum line is prevented by indirect retainers placed against definite rest seats on the opposite side of the fulcrum line. Indirect retainers should be placed as far as possible from the distal extension base to gain the best possible leverage advantage against the lifting of the distal extension base.

Most tooth-borne partial dentures do not exhibit rotational movements due to their extensive tooth support.



Although the most effective location of an indirect retainer is usually in the area of an incisor tooth, these teeth may not be strong enough to support an indirect retainer. In addition, incisors often have steep lingual inclines that cannot be favourably altered to support a rest. In such cases, the nearest canine tooth or the mesial occlusal surface of the first premolar may be the best location, even though it is not as far removed from the fulcrum line.

In the absence of indirect retainers for distal end extension dentures subjected to posterior dislodging forces, two undesirable events may take place: (1) the denture base moves away from the supporting tissues, and (2) the anterior segment of the major connector impinges upon the underlying soft tissues. This results in the transmission of destructive forces to the hard and soft tissues of the dental arch.



Forms of Indirect Retainers

1. Auxiliary occlusal rest

Usually placed on the mesial occlusal surface of a first premolar tooth when there is inadequate cingulum tooth structure on the canines, or there is poor access to the lingual surfaces of the anterior teeth.



2. Auxiliary cingulum rests

The cuspid is the most easily utilized anterior tooth for the preparation of a rest seat since the cingulum is usually more prominent than the lateral or central incisors.

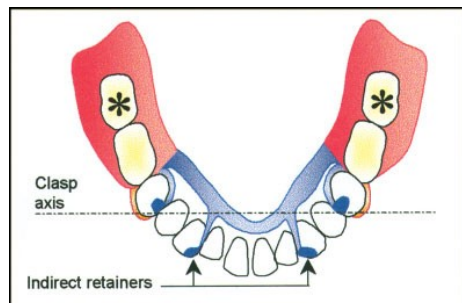
3. Canine extension from occlusal rest

A finger extension from a premolar rest is placed on the prepared lingual slope of the adjacent canine tooth when the first premolar must serve as a primary abutment.



4. Incisal rest

An incisal rest also may provide an indirect retainer where other rests are contraindicated. This is particularly true for maxillary and mandibular incisors, as well as mandibular canines. Because of the unfavourable lingual anatomy of these teeth, incisal rests may be the only acceptable option. Unfortunately, incisal rests are esthetically objectionable and exhibit long approach arms that may transfer harmful tipping forces to abutments. A better solution would be to use one of the modifications for a lingual rest on these teeth.



5. Continuous bar retainers and lingual plates

Technically, continuous bar retainers and lingual plates are not indirect retainers since they rest on unprepared lingual inclines of anterior teeth. The indirect retainers are the terminal rests at either end in the form of auxiliary occlusal rests or canine rests.



6. Modification areas

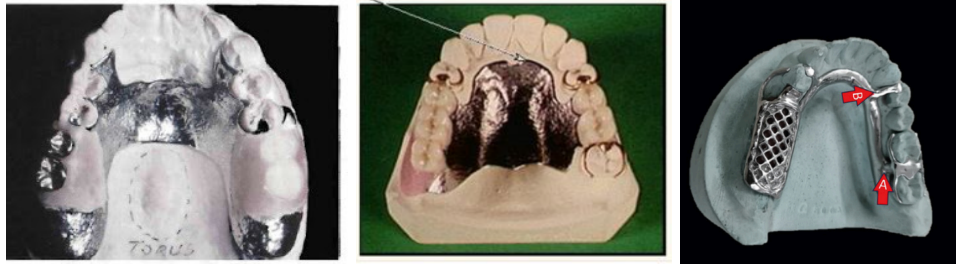
The occlusal rest on a secondary abutment in a Class II partial denture may serve as an indirect retainer. A secondary abutment is adjacent to a bounded edentulous span other than the free-end extension. If the modification space were not present, as in the unmodified Class II arch, auxiliary occlusal rest and stabilising components would be essential to the design of the denture. However, a modification space conveniently provides an abutment tooth for support, stabilization, and retention.



7. Rugae Support

The rugae area of the maxillary arch can be used as a means of indirect retention because the rugae area is firm and usually well-situated to provide indirect retention for a Class I removable partial denture. Although this is true, rugae coverage is undesirable and should be avoided if possible.

A U-shaped maxillary major connector (palatal horseshoe design) typically uses rugae support for indirect retention. Due to the lack of a posterior palatal seal, posterior retention is insufficient. The need for indirect retention necessitates the use of rugae coverage.



(A) Minor connectors that join clasp assemblies to major connectors.

(B) Minor connectors that join indirect retainers or auxiliary rests to major connectors.

